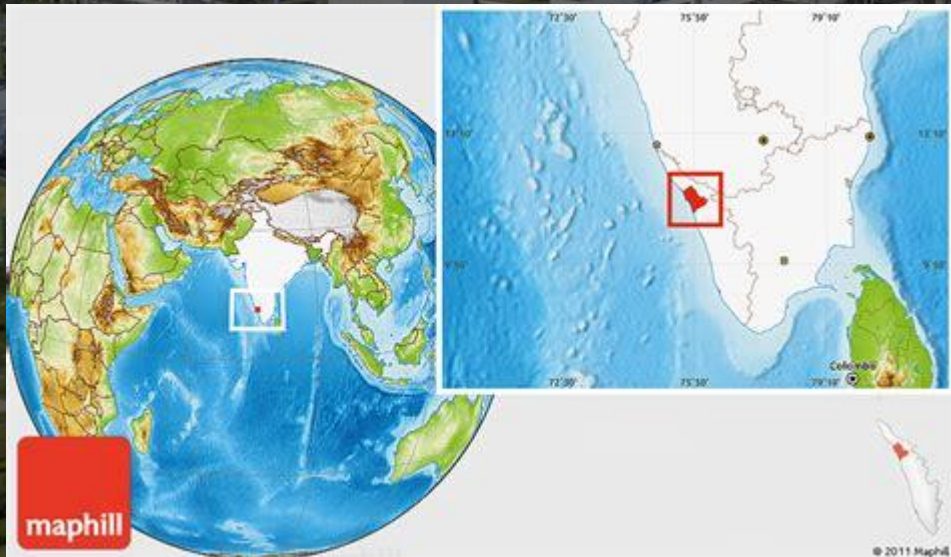




Waste to Energy

Kozhikode (India)

Group 8

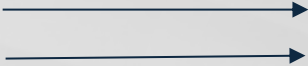




Current situation in Kozhikode



Biodegradable waste



Non-biodegradable waste, little amount due to non proper sorting

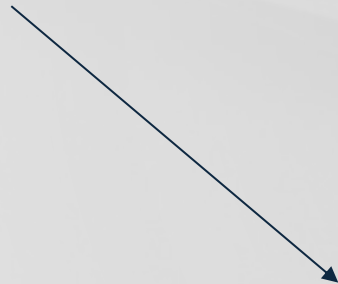


Windrow composting

(Njelliyamparamba):

- Pollution issues due to non-biodegradable waste presence
- Not enough capacity

Collection and sorting



Major part of non-biodegradable waste



Thermal power plant (128MW)

How to handle the waste collection and process in order to minimize the environmental impact?

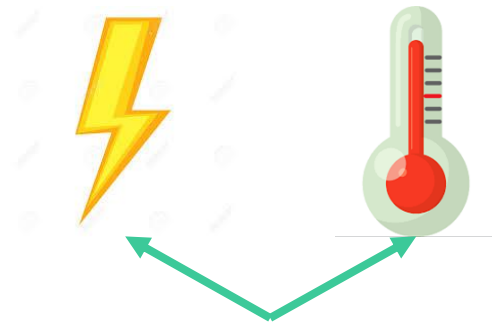


Solution

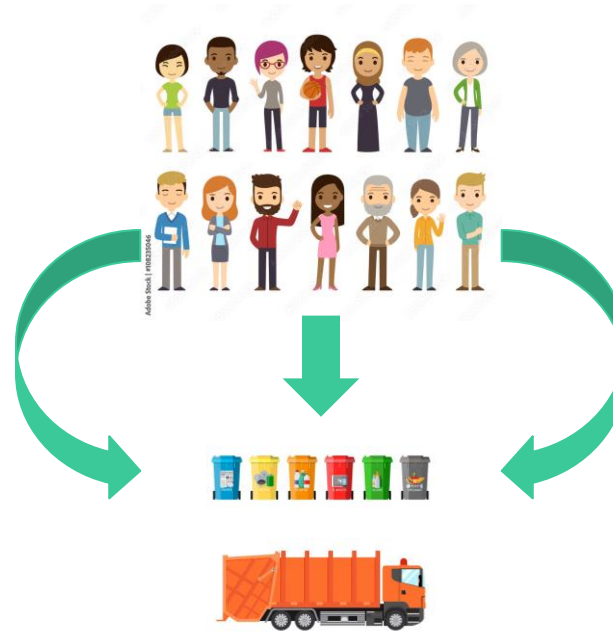
600 ton/day

Electricity and heat

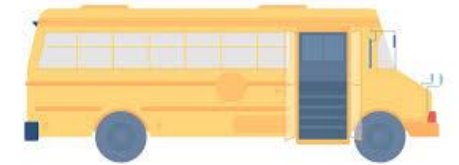
Public transport



Waste



Biological
Waste



50%/50%

Incineration

- steam power plant in cold condensing mode
- grate furnace instead of CFB furnace
- flue gas cooled below 200°C before entering the pollution control system
- 20% electrical efficiency
- Burnable waste LHV = 16 MJ/kg
- Power generated:

$$P_{burn} = \frac{300 * 10^3 * 16 * 0,2}{24 * 3600}$$

$$P_{burn} = 11,11MW$$

Can reduce the capacity of diesel power plant (128MW) already existing in Kozhikode.



Biological treatment

- 600 tonnes of water added (1:2 ratio)
- Biological waste LHV = 4 MJ/kg
- Retention time: 30 days
- Volume of the digester needed: 36 000 m³
- Total solid 20%, volatile solids 80%
- Calculations:

$$OLR = \frac{900m^3/day * 53,3kg_{VS}/m^3}{27000m^3} = 1,78kg_{VS}/m^3/day$$

$$Biogasyield = 1,78kg_{VS}/m^3/day * 0,67m^3/kg_{VS} * 27000m^3$$

$$Biogasyield = 32200,2m^3/day$$

Can power 214 biogas buses a day



Environmental benefits

- Reduced use of diesel for public transport
- Increased waste collection and improved waste management
- Reduced needs of landfills
- Reduced need of electricity production from coal power plants



Recommendation

- Construct a Waste to Energy plant ie; an MSW grate incinerator coupled with a steam power plant.
- The plant would have a capacity of 11.11 MW.
- Could replace the diesel power plant in Kozhikode
- Consider the biological treatment of biowaste by constructing a biogas plant.
- It could produce 32200.2 m³ of biogas per day which can be used to fuel 214 biogas buses.

